



Healthcare Patient Experience Analytics

Power BI Business Intelligence Project

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About NHQC & Project Context

About NHQC

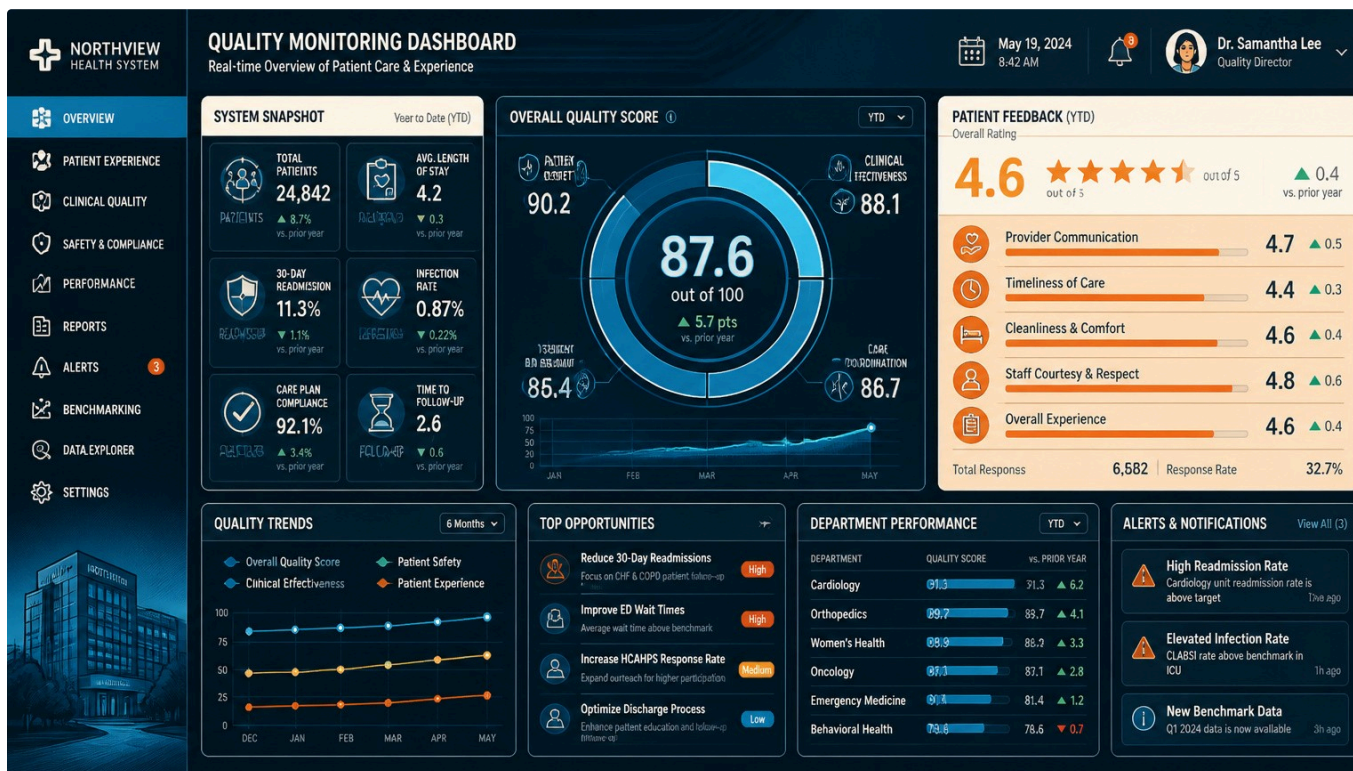
- Monitors healthcare quality nationwide using HCAHPS survey data
- Improves patient experience and supports healthcare policy
- Enables evidence-based decision making

Business Problem

Hospitals collect millions of patient surveys annually. The challenge is transforming this volume of data into actionable insights for executives.

Project Goal

Develop an interactive Power BI dashboard to monitor patient satisfaction and healthcare quality nationwide.





BUSINESS OBJECTIVES

What We Set Out to Achieve



Analyze & Benchmark

Analyze national patient satisfaction, compare states and regions, and benchmark performance against quality measures.



Monitor Trends

Track reporting period trends and build DAX KPIs for year-over-year performance measurement.



Build Dashboards

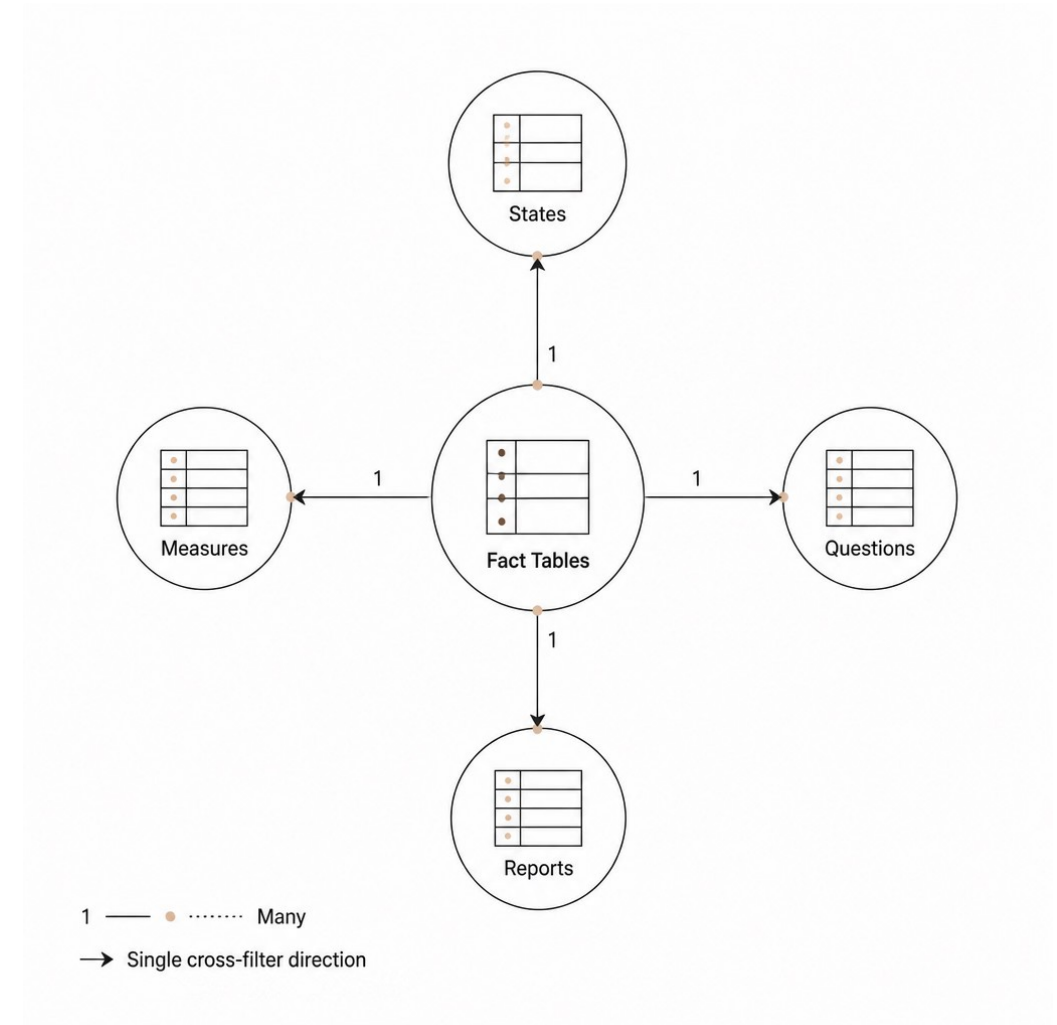
Develop interactive Power BI dashboards that support executive decision making at every level.

Dataset Overview: Star Schema Design

7 Tables • Optimized for Performance

The data model follows a **Star Schema** architecture with one-to-many relationships and single-direction cross-filtering — ensuring fast query performance and accurate reporting across all dashboards.

- **Dimension Tables (4):** States, Measures, Questions, Reports
- **Fact Tables (3):** National Results, State Results, Responses



Key Performance Indicators at a Glance

71.41%

Overall Satisfaction

National patient satisfaction rate

26.61%

Survey Response Rate

Across all participating hospitals

51

Participating States

Nationwide coverage

10

Quality Measures

Tracked in the dashboard

-1.30

Year-over-Year Change

Latest reporting period shift

✓ Highest Performing

Discharge Information

↑ Requires Attention

Care Transition

Six Interactive Dashboard Pages

Each page is designed for a specific analytical purpose — from executive summaries to granular regional comparisons.



What the Data Reveals



Regional Performance

West North Central leads the nation in patient satisfaction scores. The **Mid-Atlantic** region shows the greatest opportunity for improvement.

Trend Analysis

Patient satisfaction grew steadily until **2021**, then declined during **2022–2023** — signaling a need for targeted intervention.

Performance Gaps

State-level performance varies significantly, underscoring the value of benchmarking against top-performing states.

Strategic Actions for Healthcare Leaders



Improve Care Transition

Strengthen discharge protocols and follow-up communication.



Medication Communication

Enhance patient education on medications at discharge.



Increase Survey Participation

Boost response rates for more representative data.



Reduce Regional Gaps

Deploy targeted interventions in underperforming regions.



Benchmark Top States

Adopt best practices from highest-performing states.



Hospital Environment

Invest in facility quality to improve patient experience.



Expand KPI Monitoring

Track additional quality measures in real time.



Data-Driven Decisions

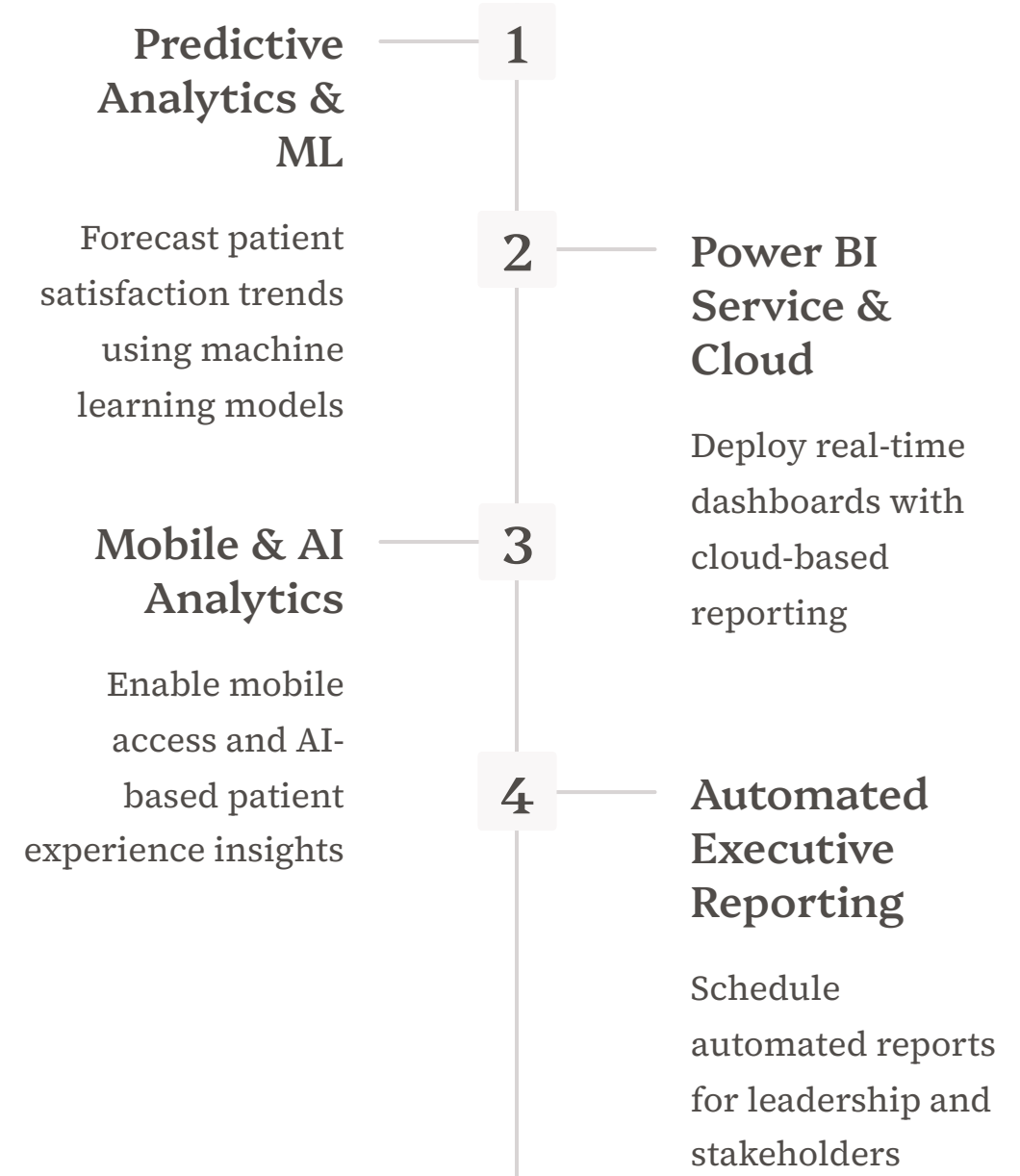
Embed analytics into executive workflows.

Value Delivered & Path Forward

Business Impact

- **Improved Patient Satisfaction** — data-driven insights enable targeted improvements
- **Better Executive Decisions** — interactive reporting replaces static reports
- **National Benchmarking** — regional comparison drives accountability
- **Evidence-Based Policy** — supports healthcare quality improvement

Future Roadmap



Project Summary & Thank You

✓ Data Model 7 Tables · Star Schema · DAX Measures	✓ Dashboards 6 Interactive Pages · KPI Framework
✓ Insights Healthcare Insights · Executive Recommendations	✓ Solution End-to-End Business Intelligence

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Questions & Discussion Welcome